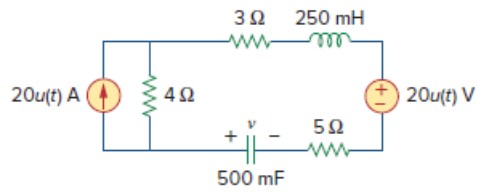


## Problem

Determine  $v$  for  $t > 0$  in the circuit in Fig. 16.60.

**Figure 16.60:**



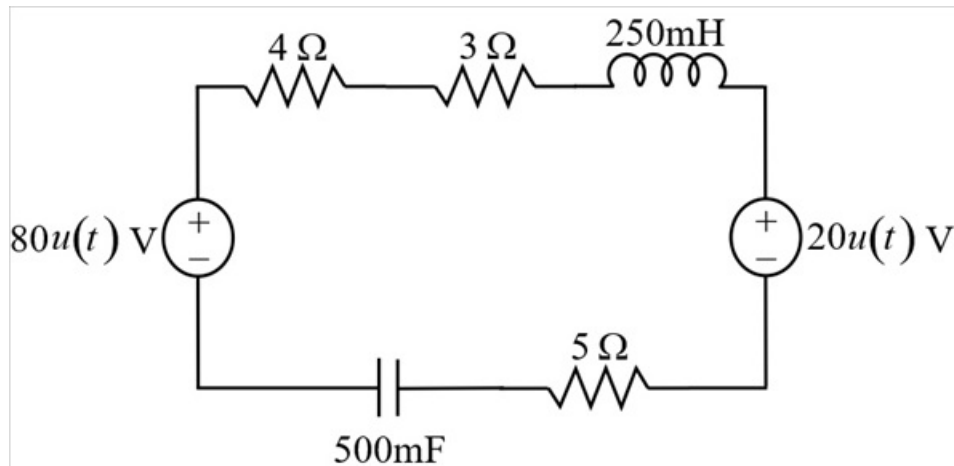
## Step-by-step solution

## Step 1 of 6

(Refer to Figure 16.60 in the textbook) The resistance  $4\ \Omega$  and current source of  $20u(t)\text{ A}$  is in parallel. Then, convert the current source in voltage source.

Put the resistance in series with voltage source and the voltage source is,

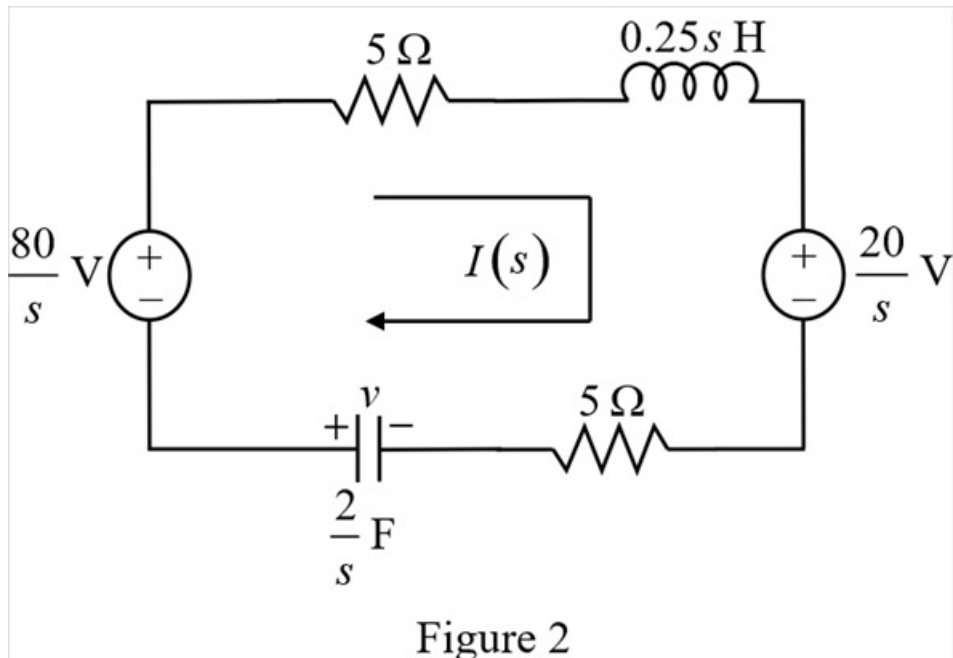
$$\begin{aligned} V &= (20u(t))(4) \\ &= 80u(t)\text{ V} \end{aligned}$$



**Figure 1**

## Step 2 of 6

Transform the Figure 1 in s-domain.



**Step 3** of 6

Apply the Kirchhoff's voltage law in the loop.

$$\frac{80}{s} = 7I(s) + 0.25sI(s) + \frac{20}{s} + 5I(s) + \frac{2}{s}I(s)$$

$$\frac{60}{s} = I(s) \left[ \frac{0.25s^2 + 12s + 2}{s} \right]$$

$$I(s) = \frac{60}{0.25s^2 + 12s + 2}$$

**Step 4** of 6

The voltage  $V(s)$  across the capacitor is,

$$V(s) = I(s) \left( \frac{2}{s} \right)$$

The flow of current is in opposite direction to the voltage drop. Then,

$$V(s) = -I(s) \left( \frac{2}{s} \right)$$

Substitute  $\frac{60}{0.25s^2 + 12s + 2}$  for  $I(s)$ .

$$\begin{aligned} V(s) &= \left( \frac{-60}{0.25s^2 + 12s + 2} \right) \left( \frac{2}{s} \right) \\ &= \frac{-120}{s(0.25s^2 + 12s + 2)} \end{aligned}$$

Rewrite the above equation.

$$\begin{aligned} V(s) &= \frac{-480}{s(s^2 + 48s + 8)} \\ &= - \left( \frac{480}{s(s + 0.167)(s + 47.84)} \right) \dots\dots (1) \end{aligned}$$

**Step 5** of 6

Calculate the partial fraction of equation (1).

$$-\left(\frac{480}{s(s+0.167)(s+47.84)}\right) = -\left(\frac{A}{s} + \frac{B}{(s+0.167)} + \frac{C}{(s+47.84)}\right) \dots\dots (2)$$

$$480 = A(s+0.167)(s+47.84) + Bs(s+47.84) + Cs(s+0.167)$$

$$480 = s^2(A+B+C) + s(48A+47.84B+0.167C) + 7.989A$$

Compare the coefficient of  $s^2$ ,  $s^1$  and  $s^0$ .

$$A = 60$$

$$B = -60.2$$

$$C = 0.2$$

**Step 6** of 6

Substitute 60 for  $A$ , -60.2 for  $B$  and 0.2 for  $C$  in equation (2).

$$-\left(\frac{480}{s(s+0.167)(s+47.84)}\right) = -\left(\frac{60}{s} - \frac{60.2}{(s+0.167)} + \frac{0.2}{(s+47.84)}\right)$$

Rewrite the above equation.

$$V(s) = -\frac{60}{s} + \frac{60.2}{(s+0.167)} - \frac{0.2}{(s+47.84)} \dots\dots (3)$$

The shifting property of inverse Laplace transform is,

$$\frac{1}{s+a} \leftrightarrow e^{-at}$$

The inverse Laplace transform of equation (3) is,

$$v(t) = (60t - 60.2e^{-0.167t} + 0.2e^{-47.84t})u(t)$$

Hence, the voltage across the capacitor is  $\boxed{(-60t + 60.2e^{-0.167t} - 0.2e^{-47.84t})u(t) \text{ V}}.$